LE CANH HAI

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PROFILE

Graduate in Mathematics and Informatics from HUST (GPA 3.89/4.00; faculty valedictorian) with research experience in optimization algorithms for variational inequalities and in nonlinear dynamical systems. Applying for a Master's in mathematics in Germany (from October 2027), aiming for doctoral research in applied mathematics.
Research Interests: Applied Mathematics; Optimization; Dynamical Systems; Mathematical Modelling.

EDUCATION

Hanoi University of Science and Technology (HUST) Sep 2022 – Aug 2026
Bachelor's Degree in Mathematics and Informatics Hanoi, Vietnam

GPA: 3.89/4.00 (**Excellent** classification); **Graduation Valedictorian** of the Faculty of Mathematics and Informatics, 2026; **ranked 1st of 166** in the Mathematics and Informatics programme
Thesis: A Self-Adaptive Inertial Algorithm for Split Variational Inequalities with Applications to Traffic Networks (9.6/10). Supervisor: Assoc. Prof. Nguyen Thi Thu Thuy

RESEARCH EXPERIENCE

COSMOS Lab, VinUniversity Sep 2026 – Present
Research Assistant, supervised by Prof. Ali Moussaoui Hanoi, Vietnam

Project: MSY and Hopf Bifurcation for a Fishery with a Rarity-Dependent Price *In preparation*

- Model fisheries with rarity-dependent prices, bilinear or saturated harvesting, and marine protected areas.
- Analyze equilibria, local and global stability, maximum sustainable yield and Hopf bifurcation thresholds.
- Derive analytical results and simulate the transition from stable equilibria to sustained oscillations.

Hanoi University of Science and Technology (HUST) Feb 2025 – Present
Research Assistant, supervised by Assoc. Prof. Nguyen Thi Thu Thuy Hanoi, Vietnam

Project: A Relaxed Golden-Ratio Extragradient Algorithm for Variational Inequalities without Monotonicity *In preparation*

- Co-developed the method, which uses alternated inertia and adaptive step sizes without a Lipschitz constant.
- Proved weak convergence via the dual variational inequality and an R-linear rate under an error bound.
- Ran numerical experiments in which the method needed fewer iterations and less CPU time than several compared algorithms, and applied it to smart-grid demand-side management.

Project: A Self-Adaptive Inertial Algorithm for Split Variational Inequalities with Applications to Traffic Networks *Conference paper*

- The algorithm needs no Lipschitz constants or operator norms, which earlier methods required in advance.
- Proved strong convergence; in numerical experiments the method needed fewer iterations and less CPU time than the compared algorithms, and was applied to traffic-network equilibrium.
- First author and presenter, Student Research Conference, HUST, May 2026.

HONORS & SCHOLARSHIPS

- **HUST Academic Achievement Scholarship**, awarded for **6 semesters** on academic results.
- **"Chap Canh Bach Khoa" Scholarship**, **7 semesters**, for outstanding students facing financial hardship.
- **University Outstanding Student**, HUST, academic year 2024–2025.
- **Outstanding Student Certificate for Academic and Research Excellence** (faculty level, HUST).

ACADEMIC & INTERNATIONAL PROGRAMS

KAIST International Summer School for Advanced Undergraduate Students Jul 2025
Fully Funded Participant Daejeon, South Korea

- Completed advanced courses in PDEs, Stochastic Analysis, Financial Mathematics, Algebra and Statistics.

VIASM Summer School on Optimization and Decision Theory Jul 2026
Participant Quy Nhon, Vietnam

- Organized by the Vietnam Institute for Advanced Study in Mathematics, with lectures by Prof. Patrick L. Combettes (NC State) on Krasnosel'skii–Mann iterations, from the fundamentals to stochastic iterations and convergence rates.

The "Red Journey": Vietnam–China Youth Cultural & Academic Exchange Nov 2025
HUST Representative Wuhan and Chongqing, China

- Represented HUST in academic and cultural exchanges with Chinese universities and youth organizations.

LEADERSHIP & COMMUNITY ENGAGEMENT

Chap Canh Bach Khoa Community 2022 – Present
Volunteer Tutor; Tutoring Program Coordinator Vietnam

- Taught and later coordinated a volunteer program giving free mathematics lessons to more than 50 secondary-school students in Mu Cang Chai, a remote mountainous area of northern Vietnam.
- As coordinator, organized volunteer tutors, schedules and communication with local schools.

TECHNICAL SKILLS & LANGUAGES

Programming Python, MATLAB, C/C++, LaTeX

English IELTS Academic 6.5 (2026)

Vietnamese Native